Prathamesh Chavan

Pune, Maharashtra

8847704787 | prathamesh1.264@gmail.com

GitHub

❖ Education

Dr. D Y Patil Institute of Engineering, Management and Reasearch, Pune **(Aug 2023 – Present)**
B.E. in Artificial Intelligence and Data Science **CGPA 8.42/10**
Higher Secondary Education (HSC-12th) **85.33%**
Secondary Education (SSC-10th) **90.80%**

❖ Skills

Languages: Python, SQL
Frameworks: Langchain, TensorFlow, PyTorch, Scikit-learn, Fast API
Databases: MySQL, MongoDB
Tools : Git, GitHub
Fundamentals: DBMS, OS, CN, DSA

❖ Internship

Codtech IT Solutions Private Limited
Data Science Intern (January 2026 – February 2026)

• Developed and automated data preprocessing and ETL pipelines using Python, Pandas, and Scikit-learn, improving data cleaning efficiency and streamlining transformation workflows for machine learning tasks
• Built and trained deep learning models for image classification and NLP-based tasks using TensorFlow/PyTorch, achieving accurate predictions with effective visualization and performance analysis
• Designed an end-to-end data science application involving data collection, preprocessing, model training, and deployment using Flask/FastAPI for real-time predictions and interactive model access

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❖ Key Projects

➢ **SalesCast Pro — AI-Powered Sales Forecasting Platform**

Tech: Python, Streamlit, SQLite, Prophet, XGBoost, SHAP, EasyOCR, Plotly

- An end-to-end sales intelligence web app using Python and Streamlit with auto-detection of CSV/Excel columns (revenue, profit, cost, quantity, category) using keyword-priority matching — eliminating manual column mapping.
- Anomaly detection engine using IQR-based statistical analysis to flag unusual sales spikes or drops, with auto-generated plain-English explanations for each anomaly.
- SHAP with an XGBoost model to provide feature-level explainability, showing which factors (day of week, seasonality, trends) drive sales predictions.
- OCR-based bill scanning using EasyOCR to extract structured data (date, amount, items) from uploaded invoice images and auto-ingest into the SQLite database for analysis.

➢ **Smart Medicine Adherence System — AI-Powered Healthcare Platform**

Tech: Python, FastAPI, Scikit-Learn, Random Forest, LangChain, EasyOCR, PostgreSQL

- A healthcare web app created with Python and FastAPI to help elderly patients track their health and take medication on time.
- Predictive health model using machine learning (Random Forest) to analyze daily patient vitals like blood pressure and blood sugar, automatically forecasting the early risk of future diseases or heart attacks.
- AI chatbot assistant powered by LangChain (RAG) to accurately answer patient questions about drug side effects and daily medication schedules.
- OCR prescription scanner using EasyOCR to read uploaded medical images and automatically save the extracted medicine names, dosages, and instructions into a PostgreSQL database.
- IoT hardware integration that wirelessly connects a custom-built smart pillbox to the web app via MQTT.

➢ **Natural Language Database Interface (SQL Agent)**

Tech: **Python, Flask, SQLite, HTML, CSS**

- Text-to-SQL engine that translates natural language questions into structured SQL queries using LLM integration.
- A robust Flask backend to interface with a SQLite database, enabling seamless real-time data retrieval.
- Schema-aware prompt engineering and a secure execution layer to ensure query accuracy and protect against unauthorized database operations.
- Streamlined data accessibility by bridging the gap between English queries and complex database schemas, allowing non-technical users to extract insights
- Implemented management capabilities within the interface including search, filter, and pagination for query results.

❖ Certifications

➢ **A.I. & Machine Learning, Data Science Bootcamp** by **Andrei Neagoie, Daniel Bourke** **(Udemy)**